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Started on Monday, March 14, 2022, 12:53 AM

State Finished

Completed on Wednesday, March 16, 2022, 8:50 PM

Time taken 2 days 19 hours

Points 23.34/30.00

Grade 77.80 out of 100.00



Question 1

Correct

10.00 points out of 10.00

Soft tissue follows an exponential deformation behavior in uniaxial tension while it is in the physiologic or normal range of elongation. This can be expressed as

$$\sigma = \frac{E_0}{a}(e^{a\epsilon} - 1)$$

where σ = stress, ϵ = strain, and E_0 and a are material constants that are determined experimentally. To evaluate the two material constants, the above equation is differentiated with respect to ϵ , which is a fundamental relationship for soft tissue

$$\frac{d\sigma}{d\epsilon} = E_0 + a\sigma$$

To evaluate E_0 and a , stress-strain data are used to plot $d\sigma/d\epsilon$ versus σ and the slope and intercept of this plot are the two material constants, respectively. The table contains stress-strain data for heart chordae tendineae (small tendons use to hold heart valves closed during contraction of the heart muscle). This is data from loading the tissue; different curves are produced on unloading.

$\sigma \times 10^3 \text{ N/m}^2$	87.8	96.6	176	263	351	571	834	1229	1624	2107	2678	3380	4258
$\epsilon \times 10^{-3} \text{ m/m}$	153	204	255	306	357	408	459	510	561	612	663	714	765

Match the σ and ϵ values to the corresponding

$$\frac{d\sigma}{d\epsilon}$$

values using finite differencing based on the given data. You can calculate all the values first and then do the matching.

1. $\sigma = 87.8$ and $\epsilon = 153$
2. $\sigma = 2677$ and $\epsilon = 663$
3. $\sigma = 350$ and $\epsilon = 357$
4. $\sigma = 1227$ and $\epsilon = 510$
5. $\sigma = 96.6$ and $\epsilon = 204$
6. $\sigma = 176$ and $\epsilon = 255$

0.1725	1	✓
0.8647	5	✓
1.6314	6	✓
3.0196	3	✓
7.7451	4	✓
12.4804	2	✓

The following relationships can be used to analyze uniform beams subject to distributed loads,

$$\frac{dy}{dx} = \theta(x); \quad \frac{d\theta}{dx} = \frac{M(x)}{EI}; \quad \frac{dM}{dx} = V(x); \quad \frac{dV}{dx} = -w(x)$$

where x = distance along beam (m), y = deflection (m), $\theta(x)$ = slope (m/m), E = modulus of elasticity (Pa = N/m²), I = moment of inertia (m⁴), $M(x)$ = moment (Nm), $V(x)$ = shear (N), and $w(x)$ = distributed load (N/m). For the case of a linearly increasing load, the slope can be computed analytically as

$$\theta(x) = \frac{w_0}{120EIL}(-5x^4 + 6L^2x^2 - L^4) \quad \dots (1)$$

Base your numerical calculations on values of the slope computed with equation (1) at equally-spaced intervals of $\Delta x = 0.125$ m along a 3.0 m beam. Use the following parameter values in your computation: $E = 200$ GPa, $I = 0.0003$ m⁴, $w_0 = 250$ kN/m. In addition, the deflections at the ends of the beam are set at $y(0) = y(L) = 0$.

Based on the above information please answer the following 3 questions.

Question 2

Correct

3.34 points out of 3.34

Question 2a

Calculate the deflection 'y' at $x = 1.5$ m using integration. Report y in centimetres. HINT: The answer is negative.

Answer:



Question 3

Incorrect

0.00 points out of 3.33

Question 2b

Calculate the moment M at $x = 1.5$ m using numerical differentiation. Report in Nm (Newton-meter).

Answer:



Question 4

Incorrect

0.00 points out of 3.33

Question 2c

Calculate the shear V at $x = 1.5$ m using numerical differentiation. Report in N (Newton). HINT: the answer is negative.

Answer: ✖

Question 5

Correct

10.00 points out of 10.00

Faraday's law characterizes the voltage drop across an inductor as

$$V_L = L \frac{di}{dt}$$

where V_L = voltage drop (V), L = inductance (in henrys; $1 \text{ H} = 1 \text{ V} \cdot \text{s/A}$), i = current (A), and t = time (s). The voltage drop can be determined as a function of time from the following data for an inductance of 4 H.

t	0	0.1	0.2	0.3	0.5	0.7
i	0	0.16	0.32	0.56	0.84	2.0

Match the time to their corresponding voltage drop.

1. $t = 0.2$
2. $t = 0.3$
3. $t = 0$
4. $t = 0.5$
5. $t = 0.7$

$V_L = 6.4$	3	✓
$V_L = 8$	1	✓
$V_L = 6.93$	2	✓
$V_L = 14.4$	4	✓
$V_L = 23.2$	5	✓

◀ Quiz 2

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Quiz 4 ►

